**5** Fig. 5.1 shows a metal sphere S. The sphere has been charged with a negative charge.

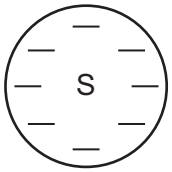


**Fig. 5.1**

**(a) (i)** There is an electric field around sphere S.

On Fig. 5.1, draw **four** field lines to show the pattern of the field and indicate the direction of the field with arrows on the lines. [2]

**(ii)** Fig. 5.2 shows a position X next to sphere S.

A small negatively charged particle is placed at position X.

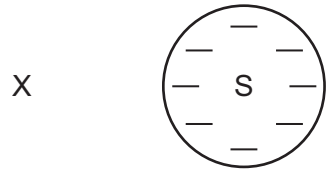


**Fig. 5.2**

State the direction of the force on the negatively charged particle at X due to the electric field around sphere S.

..... [1]

**(iii)** The negatively charged particle at X is released from rest.

Describe the motion of the small negatively charged particle due to the electric field around sphere S.

.....

.....

..... [2]

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**(b)** Fig. 5.3 shows sphere S being spray painted. Sphere S is negatively charged. As the paint particles exit the wide nozzle of the paint sprayer, they become charged with a positive charge.

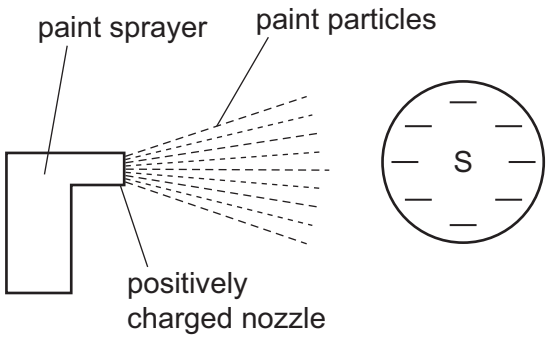


**Fig. 5.3**

**(i)** Explain why the paint particles spread out when they leave the nozzle.

.....

..... [1]

**(ii)** The sphere can be painted by hand using a paintbrush.

Suggest and explain **one** advantage to using charged paint from a spray gun to paint sphere S.

advantage .....

explanation .....

.....

.....

[2]

[Total: 8]

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